# College Event Management System - Web Application Design

## 1. OVERVIEW

A comprehensive web application for managing college events, allowing students to discover, register for, and manage event participation while enabling administrators to create and oversee events.

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## 2. USER ROLES

**Student/Participant**

- Browse and search events
- Register for events
- Manage personal profile
- View registered events
- Cancel registrations

**Event Organizer**

- Create and manage events
- View registrations
- Generate attendance reports
- Send notifications to participants

**Administrator**

- Manage all events
- Manage users
- View system analytics
- Approve/reject events

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## 3. WEB PAGES & FEATURES

### 3.1 HOME PAGE (/)

**Purpose:** Landing page showcasing featured events and quick access to main features

**Components:**

- **Hero Section**
  - Welcome banner with tagline
  - Search bar (quick event search)
  - CTA buttons: "Browse Events" | "Register Now"
- **Featured Events Carousel**
  - Top 5-6 upcoming events
  - Event cards with: thumbnail, title, date, location, category
  - "View Details" button on each card
- **Categories Section**
  - Grid of event categories (Cultural, Technical, Sports, Workshops, Seminars, etc.)
  - Icon + category name
  - Click to filter events
- **Statistics Dashboard**
  - Total events
  - Active participants
  - Upcoming events this month
  - Success stories
- **Recent Events Timeline**
  - Last 3-4 completed events with photos
  - Brief descriptions
- **Footer**
  - Quick links (About, Contact, FAQs)
  - Social media links
  - Copyright info

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### 3.2 USER REGISTRATION PAGE (`/register`)

**Purpose:** New user account creation

**Components:**

- **Registration Form**

    - Full Name*
    - Email (college email)*
    - Roll Number/Student ID*
    - Department/Branch*
    - Year of Study*
    - Phone Number*
    - Password* (with strength indicator)
    - Confirm Password*
    - Profile Picture (optional upload)
    - Terms & Conditions checkbox*
- **Validation**
    - Real-time field validation
    - Email verification link sent
    - Duplicate email/roll number check
- **Alternative Options**
    - "Already have an account? Login"
    - Social login options (optional): Google, Microsoft

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### 3.3 LOGIN PAGE (`/login`)

**Purpose:** User authentication

**Components:**

- **Login Form**
    - Email/Roll Number
    - Password
    - "Remember Me" checkbox
    - "Forgot Password?" link
- **Action Buttons**
    - Login button
    - "Don't have an account? Register" link
- **Security Features**
    - CAPTCHA (after 3 failed attempts)

- Account lockout notification

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### 3.4 USER DASHBOARD (`/dashboard`)

**Purpose:** Personalized user homepage after login

**Components:**

- **Welcome Section**
  - "Welcome back, [Name]"
  - Quick stats: Events registered, Events attended, Certificates earned
- **Upcoming Registered Events**
  - List/card view of events user registered for
  - Countdown timer for next event
  - Quick actions: View Details, Cancel Registration, Add to Calendar
- **Recommended Events**
  - Based on user's department/interests
  - "You might be interested in..." section
- **Recent Notifications**
  - Event reminders
  - Registration confirmations
  - Event updates/cancellations
- **Quick Actions Panel**
  - Browse All Events
  - View My Profile
  - My Certificates
  - Payment History

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### 3.5 USER PROFILE PAGE (`/profile`)

**Purpose:** View and edit user information

**Components:**

- **Profile Header**

    - Profile picture with edit option
    - Name, Roll Number, Department
    - Member since date
- **Personal Information Section** (Editable)
    - Full Name
    - Email
    - Phone Number
    - Department
    - Year of Study
    - Bio/About Me
- **Account Settings**
    - Change Password
    - Email Preferences (notifications)
    - Privacy Settings
- **Event History**
    - Tab 1: Registered Events
    - Tab 2: Attended Events
    - Tab 3: Past Events
- **Certificates & Achievements**
    - List of earned certificates
    - Download option for each
- **Action Buttons**
    - Save Changes
    - Cancel
    - Delete Account (with confirmation)

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### 3.6 EVENTS LISTING PAGE (`/events`)

**Purpose:** Browse all available events

**Components:**

- **Search & Filter Section**
    - Search bar (search by name, description, organizer)

    - Filters:
        - Category (Cultural, Technical, Sports, etc.)
        - Date Range (picker)
        - Location/Venue
        - Department
        - Registration Status (Open, Closed, Full)
        - Fee Type (Free, Paid)
    - Sort by: Date, Popularity, Registration Deadline
- **Events Display**
    - Grid/List view toggle
    - Event cards containing:
        - Event thumbnail/poster
        - Event title
        - Date & Time
        - Location
        - Category badge
        - Seats available
        - Registration fee (if any)
        - "Register Now" or "View Details" button
        - "Favorite" icon
- **Pagination**
    - Page numbers or infinite scroll
- **Results Count**
    - "Showing X of Y events"

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### 3.7 EVENT DETAILS PAGE (`/events/:eventId`)

**Purpose:** Comprehensive information about a specific event

**Components:**

- **Event Header**
    - Event banner/poster (large image)
    - Event title

    - Breadcrumb navigation
- **Key Information Panel**
    - Date & Time
    - Venue/Location
    - Duration
    - Category
    - Organized by
    - Registration Deadline
    - Seats: Available/Total
    - Registration Fee
- **Event Description**
    - Detailed description
    - Objectives
    - Target Audience
    - Prerequisites (if any)
- **Event Schedule/Agenda**
    - Timeline of activities
    - Speaker information (if applicable)
- **Registration Section**
    - "Register Now" button (prominent)
    - Registration status indicator
    - Terms & Conditions (if any)
- **Additional Information**
    - FAQs
    - Contact organizer
    - Event guidelines
    - Certificates info (if provided)
- **Similar Events Section**
    - 3-4 related events
- **Social Sharing**
    - Share buttons (WhatsApp, Twitter, Facebook, LinkedIn)

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### 3.8 EVENT REGISTRATION PAGE (`/events/:eventId/register`)

**Purpose:** Complete event registration process

**Components:**

- **Registration Form**
  - Pre-filled user details (Name, Email, Roll Number)
  - Additional fields (if required by event):
    - T-shirt size
    - Dietary preferences
    - Team members (for team events)
    - Special requirements
- **Payment Section** (if paid event)
  - Registration fee breakdown
  - Payment method selection (UPI, Card, Net Banking)
  - Apply discount code
- **Terms & Conditions**
  - Event-specific T&C
  - Cancellation policy
  - Acceptance checkbox
- **Confirmation**
  - Review details
  - "Confirm Registration" button
- **Progress Indicator**
  - Step 1: Details
  - Step 2: Payment (if applicable)
  - Step 3: Confirmation

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### 3.9 REGISTRATION CONFIRMATION PAGE (`/registration/success`)

**Purpose:** Confirm successful registration

**Components:**

- **Success Message**
  - Checkmark icon
  - "Registration Successful!" message
  - Registration ID/Ticket Number
- **Event Summary**
  - Event name
  - Date & Time
  - Venue
  - Registration ID
- **Next Steps**
  - Instructions for attending
  - Add to calendar button
  - Download ticket/QR code
- **Action Buttons**
  - View Event Details
  - Go to Dashboard
  - Download Receipt/Invoice (if paid)
- **Email Confirmation Notice**
  - "Confirmation email sent to [email]"

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### 3.10 MY EVENTS PAGE (`/my-events`)

**Purpose:** Manage user's event registrations

**Components:**

- **Tabs Navigation**
  - Upcoming Events
  - Past Events
  - Cancelled Events
- **Event Cards** (for each tab)
  - Event thumbnail
  - Event name

    - Date & Time
    - Venue
    - Registration ID
    - Status badge (Confirmed, Attended, Cancelled)
    - Action buttons:
        - View Details
        - Download Ticket
        - Cancel Registration (for upcoming)
        - Download Certificate (for past attended)
        - Rate Event (for past)
- **Filters**
    - Filter by category
    - Filter by date range
- **Export Options**
    - Download event list as PDF
    - Export to calendar (ICS file)

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### 3.11 SEARCH RESULTS PAGE (`/search`)

**Purpose:** Display search results

**Components:**

- **Search Query Display**
    - "Results for '[search term]'"
    - Number of results found
- **Filter Sidebar**
    - Same filters as Events Listing page
    - Clear all filters button
- **Results Section**
    - Event cards matching search
    - Highlighted search terms
    - Relevance sorting
- **No Results State**

    - Helpful message
    - Suggestions
    - Browse all events button

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### 3.12 CATEGORIES PAGE (`/categories/:categoryName`)

**Purpose:** View events in a specific category

**Components:**

- **Category Header**
    - Category icon/banner
    - Category name
    - Category description
    - Total events count
- **Events Display**
    - Similar to Events Listing page
    - Pre-filtered by category
    - Additional filters available

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### 3.13 CONTACT PAGE (`/contact`)

**Purpose:** Allow users to contact support/admin

**Components:**

- **Contact Form**
    - Name
    - Email
    - Subject
    - Message
    - Attachment (optional)
    - Submit button
- **Contact Information**
    - Email address

    - Phone number
    - Office address
    - Office hours
- **FAQ Section**
    - Common questions with answers
    - "Still need help?" CTA

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### 3.14 ABOUT PAGE (`/about`)

**Purpose:** Information about the platform

**Components:**

- **About Section**
    - Platform mission & vision
    - How it works
    - Benefits for students
- **Team Section**
    - Development team info (optional)
    - College information
- **Statistics**
    - Total events hosted
    - Active users
    - Success stories

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## 4. ORGANIZER/ADMIN PAGES

### 4.1 ORGANIZER DASHBOARD (`/organizer/dashboard`)

**Components:**

- Overview of created events
- Registration statistics
- Quick actions: Create Event, View Reports

### 4.2 CREATE EVENT PAGE (`/organizer/create-event`)

**Components:**

- **Event Details Form**
  - Event Title*
  - Description*
  - Category*
  - Date & Time*
  - Venue*
  - Duration*
  - Max Participants*
  - Registration Deadline*
  - Registration Fee (if any)
  - Event Poster/Banner Upload*
- **Additional Settings**
  - Visibility (Public/Department-specific)
  - Prerequisites
  - Certificate availability
  - Team event toggle
  - Approval required
- **Action Buttons**
  - Save as Draft
  - Publish Event
  - Preview

### 4.3 MANAGE EVENTS PAGE (`/organizer/events`)

**Components:**

- List of created events
- Status indicators (Draft, Published, Completed, Cancelled)
- Actions: Edit, Delete, View Registrations, Generate Report

### 4.4 EVENT REGISTRATIONS PAGE (`/organizer/events/:eventId/registrations`)

**Components:**

- List of registered participants
- Participant details (Name, Roll No, Email, Phone)
- Registration date & time
- Payment status (if paid)
- Actions: Export to Excel/CSV, Mark Attendance, Send Email

### 4.5 ADMIN DASHBOARD (`/admin`)

**Components:**

- System-wide analytics
- User management
- Event approval queue
- Reports & insights

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## 5. NAVIGATION STRUCTURE

**Header Navigation (All Pages)**

- Logo (links to home)
- Events
- Categories
- About
- Contact
- Search icon (opens search bar)
- Notifications bell
- User profile dropdown
    - Dashboard
    - My Events
    - Profile
    - Settings
    - Logout

**Mobile Navigation**

- Hamburger menu

- Bottom navigation bar for key features

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## 6. RESPONSIVE DESIGN CONSIDERATIONS

- Mobile-first approach
- Breakpoints: Mobile (<768px), Tablet (768-1024px), Desktop (>1024px)
- Touch-friendly buttons and spacing
- Collapsible filters on mobile
- Swipeable event cards
- Bottom sheet for filters on mobile

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## 7. KEY FEATURES ACROSS PAGES

**Common Features:**

- Loading states (spinners/skeletons)
- Error messages (toast notifications)
- Success confirmations
- Form validation (real-time)
- Breadcrumb navigation
- Back to top button

**Accessibility:**

- Alt text for images
- ARIA labels
- Keyboard navigation support
- High contrast mode
- Screen reader compatible

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## 8. SUGGESTED COLOR SCHEME & DESIGN

**Primary Colors:**

- Primary: Blue (`#2563EB`) - Trust, professionalism
- Secondary: Orange (`#F97316`) - Energy, action
- Success: Green (`#10B981`)
- Warning: Yellow (`#F59E0B`)
- Error: Red (`#EF4444`)

**Typography:**

- Headers: Bold, modern sans-serif (e.g., Inter, Poppins)
- Body: Clean, readable (e.g., Open Sans, Roboto)

**Design Style:**

- Clean and modern
- Card-based layouts
- Generous white space
- Subtle shadows and rounded corners
- Consistent iconography

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## 9. ADDITIONAL PAGES (OPTIONAL)

- `/notifications` - All notifications center
- `/certificates` - Certificate gallery
- `/calendar` - Calendar view of all events
- `/leaderboard` - Gamification (most active participants)
- `/forgot-password` - Password recovery
- `/verify-email` - Email verification
- `/terms` - Terms & Conditions
- `/privacy` - Privacy Policy
- `/help` - Help Center/Documentation

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## 10. IMPLEMENTATION RECOMMENDATIONS

**Frontend:**

- React.js / Vue.js / Angular
- Responsive CSS framework (Tailwind CSS / Bootstrap)
- State management (Redux / Vuex)

**Backend:**

- Node.js + Express / Django / Spring Boot
- RESTful API design

**Database:**

- PostgreSQL / MySQL (relational)
- MongoDB (NoSQL alternative)

**Additional Services:**

- Authentication: JWT or OAuth
- Email service: SendGrid / AWS SES
- File storage: AWS S3 / Cloudinary
- Payment gateway: Razorpay / Stripe (for paid events)

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## 11. USER FLOW EXAMPLES

**New User Registration Flow:**

1. Home page → Click "Register"
2. Fill registration form
3. Email verification
4. Login
5. Dashboard

**Event Registration Flow:**

1. Browse events or search

2. Click event card
3. View event details
4. Click "Register Now"
5. Fill registration form (+ payment if applicable)
6. Confirmation page
7. Email confirmation received

**Organizer Event Creation Flow:**

1. Organizer login
2. Organizer dashboard
3. Click "Create Event"
4. Fill event details form
5. Upload poster
6. Publish or save as draft
7. Event goes live (or pending approval)

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This design provides a complete blueprint for your college event management system. You can customize it based on your specific college requirements!